Evaluating Deontic Logic in LLMs: A Three-Layer Benchmark

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Abstract

Large language models (LLMs) are increasingly asked to reason about a plethora of subjects, a major subset being those related to obligation, permission, and prohibition. Yet whether they encode anything resembling formal (deontic) logic as opposed to imitation of normative language remains largely untested. This report introduces a three-layer benchmark that separates surface form (syntax), truth-conditional evaluation (semantics), and rule-governed derivation (proof) as distinct competencies, and reports results for four widely used models (GPT-5.5, Claude Sonnet 5, Gemini 3.6-flash, and DeepSeekV4) on 35 items with ground truth traceable to named axioms of Standard Deontic Logic (SDL), explicit Kripke countermodels, or published proofs of two classical contrary-to-duty paradoxes (Chisholm's promise paradox and Forrester's gentle-murderer paradox). All four models handle single-premise paraphrase negation-scope, and disjunction inferences almost without error. Yet three of the four models fail a majority of items when asked, whether a short contrary-to-duty statement set is internally consistent, including one item that states a contradiction directly. One model, DeepSeek V4, shows close to the reverse profile. These results give insight into the extend to which LLMs implicitly encode deontic reasoning, and how faithful it is to formal deontic logic

1 Introduction

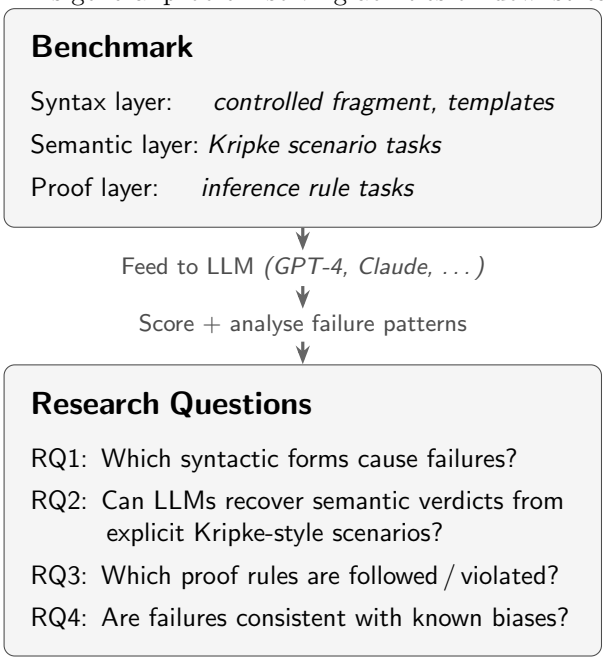
To indicate the scope of this study it is perhaps fitting to start with what will not be discussed. This paper will mainly focus on deontic logic. That is the question of what is obligatory or permitted as opposed to the question of what is logically possible (Alethic & temporal logic), what is known to be true by someone (Epistemic & doxastic) or what is desired (Boulomaic & Dynamic)

So then what is deontic logic? Deontic comes from the Greek deon, meaning duty. a deontic modal like 'may' can be interpreted as quantifying over a space of deontically possible worlds (kripke scenarios). The statement 'You may leave' is true in at least one deontically possible world where you did in fact leave. The

statement 'you must leave' quantifies universally, so that in every deontically possible world you do in fact leave.

While large language models (LLMs) have demonstrated remarkable performance across some reasoning tasks, their ability to handle deontic reasoning remains underexplored.

as to the question why this is relevant. Several studie show that LLMs performance on reasoning tasks can be highly correlated with their performance on various benchmarks[6]. Notably, Chatbot Arena Elo ratings[1], MMLU ([5]), and GSM8K ([3]), supporting the hypothesis that logical reasoning abilities aid in LLMs general problem solving abilities on downstream tasks.



2 Terminology

In this research we will mainly be focussing on these two deontic operators:

- Obligation (OA): Symbolized as OA, this represents statements such as "It is obligatory that A" or "One must do A". In a semantic Kripke scenario, this quantifies universally, meaning the action occurs in every deontically possible world.
- Permission (PA): Symbolized as PA, this represents "It is permissible that A" or "One may do A". Semantically, this indicates that the action occurs in at least one deontically possible world.

"It is obligatory that A" and "One must do A" are symbolized as OA, while permissions such as "It is permissible that A" and "One may do A" are symbolized

asPA.

3 Which Syntactic Forms Cause Failures?

This section addresses RQ1. The syntactic layer investigates whether LLM performance on deontic inference tasks varies as a function of surface form rather than logical content. All ground-truth labels are derived from Standard Deontic Logic (SDL) and independently verified against named axioms, published theorems, or explicit Kripke countermodels. No ground-truth label in this section rests on intuition alone.

Formal Background and Proof of Ground Truths

Let $\mathsf{O}A$ denote "it is obligatory that A" and $\mathsf{P}A$ denote "it is permitted that A". SDL is built on modal logic **KD** plus the following axioms [8, 14]:

N (Necessitation) If $\vdash \phi$, then $\vdash \mathsf{O}\phi$ (If something is a logical fact, it is mandatory. Logic must be obeyed.)

K (Distribution) $\mathsf{O}(\phi \to \psi) \to (\mathsf{O}\phi \to \mathsf{O}\psi)$ (If you are obligated to a result ($\phi \to \psi$) and the condition (ϕ) is met, you are obligated to the outcome (ψ))

D (Consistency) $\mathsf{O}A \to \mathsf{P}A$ (If you are required to do it, you must be allowed to do it.)

Def (Definition) $\mathsf{P}A \equiv \neg\mathsf{O}\neg A$ (A is permitted if you aren't forced to avoid it)

From **Def** alone, the four Deontic Square equivalences follow by propositional logic [8]:

$$\mathsf{O}A \leftrightarrow \neg\mathsf{P}\neg A \quad \text{(Sq-1)}$$
$$\mathsf{P}A \leftrightarrow \neg\mathsf{O}\neg A \quad \text{(Sq-2)}$$
$$\neg\mathsf{O}A \leftrightarrow \mathsf{P}\neg A \quad \text{(Sq-3)}$$
$$\neg\mathsf{P}A \leftrightarrow \mathsf{O}\neg A \quad \text{(Sq-4)}$$

These four equivalences constitute the Deontic Square and generate the surface-form tests.

Because of the four equivalences, we can test a system's "intelligence" by asking the same rule in different ways. If the system is logically sound, its answers must be consistent across every form. Critically, $\neg\mathsf{O}A$ ("not obligatory") is **not** equivalent to $\mathsf{O}\neg A$ ("obligatory not to"); confusing these two forms is a specific error the tests are designed to detect. For any human reader this would be obvious, but LLMs might struggle with such inferences.

The proof of each equivalence is a one-step application of **Def** plus double-negation elimination. For example, (Sq-3): $\neg\mathsf{O}A \leftrightarrow \neg\neg\mathsf{P}\neg A \leftrightarrow \mathsf{P}\neg A$.

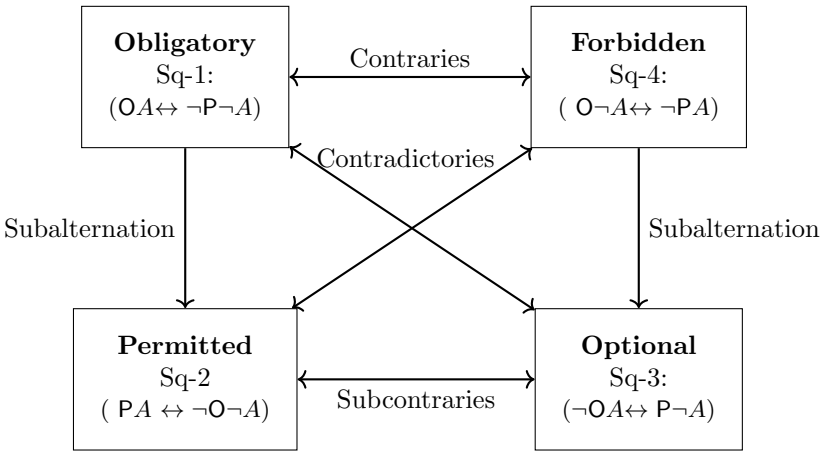


Figure 1: The Deontic Square of Opposition in SDL

For the **invalid** patterns, invalidity is shown by explicit Kripke countermodels of the form (W, R, V) where W is a set of worlds, $R \subseteq W \times W$ is the accessibility relation, and V assigns propositional truth values at each world.

Countermodel CM1. $W = \{w_1, w_2\}$; $R(w) = \{w_1, w_2\}$; $V(A, w_1) = \top$, $V(A, w_2) = \bot$.
$\mathsf{P}A$ is *true* ($w_1 \in R(w)$ and $w_1 \models A$) but $\mathsf{O}A$ is *false* ($w_2 \in R(w)$ and $w_2 \not\models A$). Hence $\mathsf{P}A \not\Rightarrow \mathsf{O}A$. □

Countermodel CM2. $W = \{w_1, w_2\}$; $R(w) = \{w_1, w_2\}$; $V(A, w_1) = \top$, $V(A, w_2) = \bot$.
$\neg\mathsf{O}A$ is *true* ($w_2 \not\models A$) but $\mathsf{O}\neg A$ is *false* ($w_1 \not\models \neg A$). Hence $\neg\mathsf{O}A \not\Rightarrow \mathsf{O}\neg A$. □

Countermodel CM3. $W = \{w_1, w_2\}$; $R(w) = \{w_1\}$; $V(A, w_1) = \bot$, $V(B, w_1) = \top$.
$\mathsf{P}(A \vee B)$ is *true* ($w_1 \models B$, so $w_1 \models A \vee B$) but $\mathsf{P}A$ is *false* (w_1 is the only accessible world and $w_1 \not\models A$). Hence $\mathsf{P}(A \vee B) \not\Rightarrow \mathsf{P}A$ [7]. □

All three countermodels contain exactly two worlds and are verifiable by inspection.

Ross's Paradox ($\mathsf{O}A \Rightarrow \mathsf{O}(A \vee B)$) is valid in SDL by the following two-step proof: (1) $A \rightarrow (A \vee B)$ is a propositional tautology, so by **N**: $\vdash \mathsf{O}(A \rightarrow (A \vee B))$. (2) By **K**: $\mathsf{O}(A \rightarrow (A \vee B)) \rightarrow (\mathsf{O}A \rightarrow \mathsf{O}(A \vee B))$; combining with step (1) yields $\mathsf{O}A \rightarrow \mathsf{O}(A \vee B)$ [12, 8]. This proof is the ground truth referenced by Test 4 below and discussed again, alongside Free Choice, in Section **??**.

Reading the tables. Each row gives a Premise-Hypothesis pair. **GT** is the ground truth: Y (inference is valid) or N (inference invalid). **Proof** names

the axiom, equation, or countermodel that proves GT. **F%** lists the failure percentage on this task. **Which?** indicates which LLM(s) were incorrect

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Test 1: Paraphrase Consistency

Rationale. All five items instantiate $\mathsf{O}A \Rightarrow \mathsf{P}A$ (Axiom D) using different vocabulary for obligation and permission. If the answers differ across items, this would indicate surface-form sensitivity.

ID	Premise	Hypothesis	Proof	GT	F%	Which?
S1-a	*You must study modal logic.*	*You are permitted to study modal logic.*	Axiom D	**Y**	0%	None
S1-b	*It is obligatory that you study modal logic.*	*It is permissible for you to study modal logic.*	Axiom D	**Y**	0%	None
S1-c	*You are required to study modal logic.*	*You may study modal logic.*	Axiom D	**Y**	0%	None
S1-d	*It is not permitted that you not study modal logic.*	*You are allowed to study modal logic.*	Axiom D + Sq-1	**Y**	0%	None
S1-e	*You cannot avoid studying modal logic.*	*studying modal logic is an option available to you.*	Axiom D + Sq-1	**Y**	0%	None

All tested models answered correctly on every S1 item, this indicates that surface vocabulary for obligation and permission (*must*, *obligatory*, *required*, *not permitted that not*, *cannot avoid*) does not in itself cause failures when the logical relationship is Axiom D applied to a single positive proposition. However, this result does not establish that models reason from Axiom D as a formal rule. It does not rule out that models have absorbed the statistical co-occurrence of obligation and permission vocabulary in natural language training data. The stronger test of formal competence is whether this paraphrase robustness persists when negation, operator inversion, or disjunction are introduced (Tests 2–4 below), and, more directly still, when the vocabulary itself is stripped of meaning.

Test 2: Negation Scope

Rationale. Items S2-a/b test $\neg \mathsf{O} A \Rightarrow \mathsf{P} \neg A$ (valid by Sq-3 (Sq-3)). Items S2-c/d/e test $\neg \mathsf{O} A \Rightarrow \mathsf{O} \neg A$ (invalid; proved by CM2). Content is held constant across the paired items.

ID	Premise	Hypothesis	Proof	GT	F%	Which?
S2-a	*It is not mandatory to brush your teeth every day.*	*It is acceptable not to take brush your teeth every day.*	Sq-3 (Sq-3)	**Y**	0%	None
S2-b	*You are not required to make your bed.*	*It is permitted not to make your bed.*	Sq-3	**Y**	0%	None
S2-c	*It is not mandatory to brush your teeth every day.*	*It is mandatory not to brush your teeth every day.*	CM2	**N**	0%	None
S2-d	*You are not required to make your bed.*	*You must not make your bed*	CM2	**N**	0%	None
S2-e	*There is no obligation to visit your parents.*	*Visiting your parents is forbidden.*	CM2	**N**	0%	None

Across the items in Test 2, the tested models recorded a 0% failure rate. The outputs aligned with the ground truth labels, distinguishing the valid inferences ($\neg \mathsf{O} A \Rightarrow \mathsf{P} \neg A$) from the invalid ones ($\neg \mathsf{O} A \Rightarrow \mathsf{O} \neg A$). For these specific single-premise templates, the data shows no evidence of the anticipated matching bias, meaning the outputs did not collapse the absence of an obligation ($\neg \mathsf{O}$) into a prohibition($\mathsf{O} \neg$).

Test 3: Operator Inversion

Rationale. Axiom D licenses $\mathsf{O} A \Rightarrow \mathsf{P} A$ but *not* $\mathsf{P} A \Rightarrow \mathsf{O} A$ (refuted by CM1). S3-f additionally combines inversion with negated content ($\mathsf{P} \neg A \Rightarrow \mathsf{O} \neg A$; refuted by CM1 with A re-labelled to $\neg A$)

ID	Premise	Hypothesis	Proof	GT	F%	Which?
S3-a	*You must water the plants.*	*You are permitted to water the plants.*	Axiom D	**Y**	25%	Deepseek
S3-b	*Drivers must follow laws.*	*Drivers may follow laws.*	Axiom D	**Y**	25%	Deepseek
S3-c	*It is permissible to smoke.*	*You are required to smoke.*	CM1	**N**	0%	None
S3-d	*You may drink.*	*You must drink.*	CM1	**N**	0%	None
S3-e	*Protesting is allowed.*	*Protesting is compulsory.*	CM1	**N**	0%	None
S3-f	*Allowed not to drive.*	*You must not drive.*	CM1	**N**	0%	None

Test 3 introduces the first errors in the dataset, with Deepseek V4 failing on S3-a and S3-b for a 25% failure rate, though it corrected its output when using its

chain-of-thought option. on the other items the models correctly predicted the correct answer, identifying patterns that represent how permission does not imply obligation. Within the scope of this study, these results show that while the models resist invalidly reversing these operators, their application of baseline deontic rules like Axiom D is fragile and sensitive to phrasing, given that they passed these same inferences under the different vocabulary in Test 1. This inconsistency, along with the chain-of-thought correction, suggests that the systems do not rely on a stable, implicitly encoded formal deontic logic, but rather on surface-level pattern matching or specific step-by-step decoding paths to navigate these boundaries

3.1 Test 4: Ross's Paradox and Free Choice

Rationale. S4-a/b/c test Ross's Paradox, SDL-valid by the N+K proof in Section 3 and confirmed in [9] S4-d/e/f test the Free Choice inference, SDL-invalid by CM3 and confirmed in [9].

ID	Premise	Hypothesis	Proof	GT	F%	Which?
S4-a	*You must groom your dog.*	*You must groom your dog or burn it.*	N+K	**Y**	0%	None
S4-b	*Required to tell the truth.*	*Required to tell the truth or the world ends.*	N+K	**Y**	0%	None
S4-c	*Obligatory to hand in my paper.*	*Obligatory to hand it in or eat it.*	N+K	**Y**	0%	None
S4-d	*May study at VU or UvA.*	*You may study at VU.*	CM3	**N**	0%	None
S4-e	*Permitted to eat A or B.*	*Permitted to eat A.*	CM3	**N**	0%	None
S4-f	*Allowed pet cat or dog.*	*Allowed to pet dog.*	CM3	**N**	0%	None

For S4-a/b/c the SDL-correct answer is Y but human intuition says N; for S4-d/e/f the SDL-correct answer is N but human intuition says Y. This test therefore looks at if the mistakes LLMs make are aligned with common mistakes in reasoning humans make because of they tend to assume that adding an option B implies that B is also a permissible way to satisfy the obligation. If the model answers Y to Ross's Paradox, it is following the formal math of SDL. If it answers N, it is following the patterns of human conversation found in its training data.

In Test 4, all tested models recorded a 0% failure rate across both the Ross's Paradox items (S4-a/b/c) and the Free Choice inferences (S4-d/e/f). These results demonstrate that, for these specific single-premise disjunction templates, the models' outputs consistently matched the formal constraints of Standard Deontic Logic over the pragmatic communication patterns found in humans.

3.2 Test 5: Contrary-to-Duty Reasoning

Rationale. Tests 1-4 test inferences from a single premise. Test 5 tests contrary-to-duty (CTD) reasoning: multi-premise scenarios in which at least one obligation is conditional on a primary obligation having already been violated. Such scenarios are often reported as prone to expose a structural limitation of SDL [2, 11]. Two published

paradoxes are used as test material, both with formally proven ground truths. All labels in this test are modified text from various proofs in the cited literature.

Chisholm's Paradox. [2] introduced the following set of four sentences, here stated as the *promise scenario* [11]:

C1. You ought to keep your promise. $\mathsf{O}k$

C2. It ought to be that if you keep your promise, you do not apologise .$\mathsf{O}(k \to \neg a)$

C3. If you do not keep your promise, you ought to apologise $\neg k \to \mathsf{O}a$

C4. You do not keep your promise. $\neg k$

Intuitively C1–C4 are consistent: primary duty (1), normal behaviour (C), contrary-to-duty remedy (3), and a statement of fact (4). However, [11] prove that the SDL formalisation for this is inconsistent:

From C1 ($\mathsf{O}k$) and C2 ($\mathsf{O}(k \to \neg a)$), by Axiom K ($\mathsf{O}(\phi \to \psi) \to (\mathsf{O}\phi \to \mathsf{O}\psi)$): $\mathsf{O}\neg a$. From C3 ($\neg k \to \mathsf{O}a$) and C4 ($\neg k$), by Modus Ponens: $\mathsf{O}a$. Hence $\mathsf{O}a \wedge \mathsf{O}\neg a$, which contradicts Axiom D ($\neg(\mathsf{O}A \wedge \mathsf{O}\neg A)$). □

Forrester's Paradox (Gentle Murderer). This paradox introduced a showing a similar scenario. Let k = "John kills Jack" and g = "Smith kills Jack gently". Note that $g \to k$ is a logical truth (killing gently implies killing).

F1. John ought not to kill Jack. $\mathsf{O}\neg k$

F2. If John kills Jack, he ought to do so gently. $k \to \mathsf{O}g$

F3. John kills Jack. k

Our college at the VU [13] prove that in SDL, F1–F3 derive $\mathsf{O}\neg k \wedge \mathsf{O}k$:

From F2 ($k \to \mathsf{O}g$) and F3 (k), by Modus Ponens: $\mathsf{O}g$. Since $g \to k$ is a tautology, by Axiom N: $\mathsf{O}(g \to k)$. By Axiom K: $\mathsf{O}g \to \mathsf{O}k$. Hence $\mathsf{O}k$. But F1 gives $\mathsf{O}\neg k$, so $\mathsf{O}k \wedge \mathsf{O}\neg k$, contradicting Axiom D.

These two proofs constitute the ground truth for all items in Test 5. The proofs are reproduced in many major survey of deontic logic and require only Axioms K, D. [8, 11].

Test Items

Test 5 is meant to test whether LLMs can correctly track the consequences of CTD sets, and whether they detect the SDL-internal contradiction. Items are grouped into three tasks: consequence identification, consistency judgement, and detachment.

Task A — Consequence identification (Chisholm).
Context given to the LLM:

> Consider the following four statements, all assumed to be true at the same time:
>
> (C1) You ought to keep your promise.
> (C2) If you keep your promise, you ought not to apologise.
> (C3) If you do not keep your promise, you ought to apologise.
> (C4) You do not keep your promise.

ID	Premise	Hypothesis	Proof	GT	F%	Which?
CA-1	*(C1) You ought to keep your promise. (C2) If you keep your promise, you ought not to apologise.*	*You ought not to apologise.*	Axiom K on C1+C2	**Y**	25%	ChatGPT
CA-2	*(C3) If you do not keep your promise, you ought to apologise. (C4) You do not keep your promise.*	*You ought to apologise.*	MP on C3+C4:	**Y**	25%	Claude
CA-3	*(C1)–(C4) all hold simultaneously. From CA-1: you ought not to apologise. From CA-2: you ought to apologise.*	*You both ought to apologise and ought not to apologise.*	CA-1 $\wedge$ CA-2: $\mathsf{O}a \wedge \mathsf{O}\neg a$; [11]	**Y**	0%	
CA-4	*A logical system simultaneously requires that you apologise and requires that you not apologise.*	*This system is logically consistent.*	Axiom D: $\neg(\mathsf{O}A \wedge \mathsf{O}\neg A)$; [8]	**N**	0%	
CA-5	*(C1)–(C4) are interpreted in Standard Deontic Logic. CA-3 holds. CA-4 holds.*	*The set C1–C4 leads to a logical contradiction in SDL.*	CA-3 + CA-4; [11]	**Y**	0%	

CA-1 and CA-2 each require a single inference step and are likely answered correctly. CA-3 requires the model to conjoin two separately derived obligations, which conflicts with the pragmatic expectation that contradictions should be avoided. Most models are expected to answer **N** to CA-3, because generating a contradiction feels like a failure. CA-4 and CA-5 then complete the paradox: the correct answer to CA-5 is **Y** — the set *is* contradictory in SDL — but LLMs are expected to answer **N**, resolving the tension by trusting natural language coherence over formal derivation.

he experimental data reveals an unexpected trend: the tested models recorded a 0% failure rate on the complex tracking items CA-3, CA-4, and CA-5. but vulnerability appeared instead in the baseline, single-step inferences. ChatGPT failed the primary deduction in CA-1, and Claude failed the basic detachment step in CA-2, with both errors resulting in a 25% failure rate on those rows. Ultimately, these findings suggest that while LLMs can at times formal contradictions when guided through explicit, sequential steps, individual localized inferences within multi-premise contexts remain fragile and susceptible to sporadic failures, it hints at the stochastic nature of LLMs and how reasoning is inherently different from what we see in human cognition.

Task B — Consequence identification (Forrester).
Context given to the model:

> Assume the following are all true at the same time.
>
> (F1) Smith ought not to kill Jones.
> (F2) If Smith kills Jones, then he ought to do so gently.
> (F3) Smith kills Jones.
> (F4) Killing someone gently logically implies killing them.

ID	Premise	Hypothesis	Proof	GT	F%	Which?
FB-1	*(F2) If Smith kills Jones, he ought to do so gently. (F3) Smith kills Jones.*	*Smith ought to kill Jones gently.*	MP on F2+F3: $\mathsf{O}g$; [4, 13]	**Y**	0%	
FB-2	*(F4) Killing gently implies killing. (FB-1) Smith ought to kill Jones gently. In SDL, obligations transfer across logical entailments.*	*Smith ought to kill Jones.*	Axiom N+K: $g \to k \Rightarrow \mathsf{O}g \to \mathsf{O}k$; [10]	**Y**	25%	Claude
FB-3	*(F1) Smith ought not to kill Jones. (FB-2) Smith ought to kill Jones.*	*These two obligations are jointly satisfiable in SDL.*	Axiom D: $\neg(\mathsf{O}k \wedge \mathsf{O}\neg k)$; [8]	**N**	0%	
FB-4	*(F1)–(F4) all hold simultaneously. FB-2 derives $\mathsf{O}k$. F1 states $\mathsf{O}\neg k$.*	*The set F1–F4 constitutes a contradiction in Standard Deontic Logic.*	Full Forrester proof; [4, 10]	**Y**	25%	ChatGPT

Failure pattern. FB-1 is a single Modus Ponens step and expected to be answered correctly. FB-2 tests consequential closurethe step that [10] identifies as the structural cause of the paradox. Most models are expected to answer **N** here, because "Smith ought to kill" conflicts strongly with world knowledge. This is precisely what we see in some of the test. Claude has difficulties identifying the correct answer for FB-2.

Task C — Consistency judgement (both paradoxes).
No additional context. Each item is self-contained.

ID	Statement set — are these jointly consistent?	Proof basis	GT	F%.	Which?
CC-1	ought to keep your promise. If you keep it, you ought not to apologise. If you break it, you ought to apologise. You break it.	Chisholm proof: SDL-inconsistent; [11]	N	75%	ChatGPT,Gemini,Claude
CC-2	ought not to kill. If you kill, you ought to kill gently. You kill. Killing gently implies killing.	Forrester proof: SDL-inconsistent; [4]	N	75%	ChatGPT,Gemini,Claude
CC-3	ought to pay your taxes. You are permitted to pay your taxes.	Axiom D; trivially SDL-consistent; [8]	Y	0%	None
CC-4	ought to submit the report. You are forbidden to submit the report.	$\mathsf{O}A \wedge \mathsf{O}\neg A$, contradicts D; [8]	N	75%	ChatGPT,Gemini,Claude

CC-3 and CC-4 are *control items*: CC-3 is an obviously consistent pair (expected: Y), and CC-4 is an explicit contradiction stated directly (expected: N, trivially). CC-1 and CC-2 are the hard items: they appear intuitively consistent in natural language but are formally inconsistent in SDL. The models answering yes to CC-1 or CC-2 is privileging natural language coherence over formal derivation. an overwelming majority of the models performed very bad on this test with only deepseek achieving a 100% succesrate. A insightfull finding considering Deepseek failed on other tasks which which had perfect sucessrate for the three other models.

Test 5 Summary

Table 6: Test 5: CTD tasks, proof sources, and expected failure modes

Task	Items	Paradox	Proof of GT	Expected LLM error
A	5	Chisholm	Axiom K+D+MP; [11]	Refuses to derive $\mathsf{O}a \wedge \mathsf{O}\neg a$ (CA-3)
B	4	Forrester	Axiom K+D+MP; [10]	Rejects consequential closure (FB-2)
C	4	Both + controls	D; full proofs	Judges CC-1/CC-2 consistent (world-knowledge bias)

3.3 Battery Summary

Table 7: Syntax layer batteries: proof sources and expected errors

Battery	n	Phenomenon	Proof of GT	Expected error
S1	5	Paraphrase consistency	Axiom D; Sq-1	Answers diverge across phrasings
S2	5	Negation scope	Sq-3; CM2	Conflates $\neg\mathsf{O}$ with $\mathsf{O}\neg$
S3	6	Operator inversion	Axiom D; CM1	Accepts $\mathsf{P}A \Rightarrow \mathsf{O}A$

4 Can LLMs recover semantic verdicts from explicit Kripke-style scenarios?

Because LLMs are trained on a huge corpus of natural language, there is a considerable risk for LLM just reasoning through implicit world knowledge. It is for example to be expected that any LLM could answers logic in the style 'if p then q' just because it is familiar with every version of these letters being used for logics, to stop LLM judging an inference based on world knowledge we also test on nonsense predicates.
Context given to the model:

> Consider three possible situations: w_1, w_2, w_3. From the current situation, w_1 and w_2 are acceptable (deontically accessible); w_3 is not accessible. In w_1: the student submits the report (A) and cites their sources (C). In w_2: the student submits the report (A) and does not cite their sources ($\neg C$). In w_3: the student does not submit the report ($\neg A$) and does not cite sources ($\neg C$).

ID	Question	Computation from (W, R, V)	GT	F%	Which?
K1-a	Is submitting the report (A) obligatory?	A holds in w_1 and w_2, the only accessible worlds $\Rightarrow$ $\mathsf{O}A$ true	**Y**	25%	Gemini
K1-b	Is citing sources (C) obligatory?	C holds in w_1 but not w_2, both accessible $\Rightarrow$ $\mathsf{O}C$ false	**N**	0%	
K1-c	Is citing sources (C) permitted?	C holds in w_1, an accessible world $\Rightarrow$ $\mathsf{P}C$ true	**Y**	0%	
K1-d	Is not submitting the report ($\neg A$) permitted?	$\neg A$ holds only in w_3, not accessible $\Rightarrow$ $\mathsf{P}\neg A$ false	**N**	0%	

K1-a and K1-c have the same ground truth (Y) as several S1 items, but here nothing about "submitting an report" or "citing sources" carries any normative charge on its own; the verdict follows only from which worlds are marked accessible.

4.1 Nonsense-Predicate Structural Tests

Rationale. A complementary strategy for isolating structural competence from world knowledge is to keep the logical connectives and, where relevant, the deontic operators, while replacing every content word with an invented, meaningless predicate. K2-a is a non-deontic control using bare Modus Tollens; K2-b and K2-c repeat the Axiom D and CM1 patterns already tested with ordinary vocabulary in Tests 1 and 3, but with nonsense predicates, directly testing the caveat raised at the end of Test 1 (Section 3): that 0% failure on Axiom D with real vocabulary is equally consistent with genuine rule use and with reproducing a frequent collocation.

ID	Premise	Hypothesis	GT	F%	Which?
K2-a	*If the blewble was blopped, then the flig was zandered. The flig was not zandered.*	*The blewble was not blopped.*	**Y**	0%	
K2-b	*It is obligatory that the djuck is eyered.*	*It is permitted that the djuck is eyered.*	**Y**	0%	
K2-c	*It is permitted that the pefit is sagiloped.*	*It is obligatory that the pefit is sagiloped.*	**N**	0%	

All four models answered every K2 item correctly, a 0% failure rate, even though the deontic items K2-b and K2-c use invented predicates with no prior obligation/permission association to lean on. Since K2-a confirms the models can still perform basic propositional inference under nonsense predicates, the clean result on K2-b/c speaks directly to the caveat raised at the end of Test 1 (Section 3): with the familiar *must*/*may* vocabulary removed, the models still tracked Axiom D and correctly withheld the invalid $\mathsf{P}A \Rightarrow \mathsf{O}A$ inference, tipping the balance toward genuine rule application over collocation.

5 Are failures consistent?

This section adresses RQ4, adressing whether the failures recorded in Sections 3 and 4 form a few clear biases, as is well documented for humans, or if they are better described as being idiosyncratic to each model. The strongest pattern in the data is not any single named bias but the absence of a shared failure profile across models. DeepSeek is the only model to fail the elementary Axiom D items in Test 3 (S3-a/b), a test that every other model passed without error, yet DeepSeek was the only model to pass all four Task C consistency-judgement items, where ChatGPT, Gemini, and Claude each failed three of four. If a single bias mechanism were driving these results, the same models should struggle in the same places. They do not: failures are only partially consistent with the named biases surveyed here, and are better characterised model-by-model than benchmark-wide. What best explains DeepSeek's inverted profile relative to the other three is not a shared, human-like reasoning bias inherited from training data, but something more unique to that model's own training or decoding behaviour precisely the kind of failure a benchmark-wide bias account would not predict.

6 discussion

It is important to keep in mind that LLMs are rapidly changing, therefore any claims on the performance of LLMs are subject to change in the near future when limitations are further pushed. Therefore this report should serve as a guide towards the 'reasoning' capabilities these models currently possess. Some of the limitions discussed today will proof insurmountable, some will perhaps require a paradigm shift to even realize and other might be for all intents and purposes a purely human phenomenon.

This study mapped how LLMs behave during various Task, but the "why" remains open. Future research should conduct mechanistic analyses to determine if these reasoning patterns stem from model architecture, training data, or prior exposure to similar logic puzzles.

Finally, because LLMs are continuously evolving, our findings may not generalize to future models with improved reasoning mechanisms. Model performance can change with updates to training data, architecture, and fine tuning strategies. Moreover, differences in fine tuning, prompt engineering, and instruction following abilities may contribute to performance variation, which makes it challenging to isolate the effects of model architecture alone.

7 Conclusion

Tables 10 and 11 summarise every failure recorded across Sections 3 and **??** All figures are computed directly from the F% and Which? columns reported in the preceding tables.

Table 10: Aggregate failure rate by battery (four models per item)

Battery	**Items**	**Failed / total judgements**	**Failure rate**
S1 (paraphrase)	5	0/20	0.0%
S2 (negation scope)	5	0/20	0.0%
S3 (operator inversion)	6	2/24	8.3%
S4 (Ross / Free Choice)	6	0/24	0.0%
Task A (Chisholm)	5	2/20	10.0%
Task B (Forrester)	4	2/16	12.5%
Task C (consistency)	4	9/16	56.3%
Total	**35**	**15/140**	**10.7%**

Table 11: Failure count and accuracy by model (35 items each)

Model	**Failures**	**Accuracy**
ChatGPT	5	85.7%
Claude	5	85.7%
Gemini	3	91.4%
DeepSeek V4	2	94.3%

Answering the Research Questions

RQ1 (syntax). Across 22 single-premise items spanning paraphrase (S1), negation scope (S2), operator inversion (S3), and disjunction (S4), three of the four models (ChatGPT, Gemini, and Claude) produced no errors, with DeepSeek being the only model to fail, and did so only on the two Axiom D items phrased. Robustness to paraphrase and negation is therefore high for this item set, but this alone under-determines whether the models are applying Axiom D as a rule or reproducing a frequent collocation, since every S1/S3 item pairs semantically related, real-world vocabulary; Battery K2's nonsense-predicate items (Section **??**) are designed to settle this once tested.

RQ3 (proof rules). The picture is more layered than a single accuracy figure suggests. When a CTD task is tested with its intermediate steps supplied (Tasks A and

B), the conjunction/contradiction-recognition steps (CA-3 through CA-5, FB-1, FB-3) were answered correctly by all four models; the errors that did occur (CA-1, CA-2, FB-2, FB-4) were single, scattered single-step slips, one model per item, rather than a shared blind spot. When the same underlying CTD sets are instead presented as a bare, self-contained statement list with no scaffolding (Task C), three of the four models fail three of four items, including one item (CC-4) that states a contradiction directly with no CTD structure at all. This gap between scaffolded and unscaffolded performance is the paper's most striking finding and is discussed further below.

RQ2 (semantics). Axiom D and CM1 hold up even when the familiar *must*/*may* vocabulary is replaced with (invented) nonesense predicates carrying no prior normative association. Battery K1, which presents an explicit (W, R, V) Kripke structure and asks the model to compute rather than recall the correct verdict, returned a near-perfect 6.25% failure rate (1 of 16 judgements): Gemini alone answered item K1-a incorrectly, while every other item was answered correctly by every model. Taken together, the semantic layer shows that (for this item set) LLMs can recover deontic verdicts from an explicitly stated formal structure, and that the earlier syntax-layer successes on Axiom D (Tests 1 and 3) do not reduce entirely to reproducing a frequent real-world collocation.

RQ4 (consistency of failures). Failures are not explained by a single bias uniformly affecting all models. DeepSeek V4 and the other three models show close to an inverted profile: DeepSeek alone fails the simplest syntax-layer items while being the only model to pass the hardest proof-layer task, and the reverse holds for the other three. This is more consistent with model-specific idiosyncrasy in training or decoding than with a shared, human-like reasoning bias, though the isolated content-driven failure on FB-2 (Claude rejecting "John ought to kill Jack") is consistent with a classical content effect operating in at least that one instance.

The Task C Finding

The single most important quantitative result in this study is the collapse observed in Task C (Section **??**). CC-3 and CC-4 are *control items*: CC-3 is an obviously consistent pair (expected: Y), and CC-4 is an explicit contradiction stated directly (expected: N, trivially, with no contrary-to-duty structure to reason through at all). CC-1 and CC-2 are the hard items: each appears intuitively consistent in natural language but is formally inconsistent once the Chisholm or Forrester derivation is carried out in SDL. A model that answers Y to CC-1 or CC-2 is privileging natural-language coherence over formal derivation.

An overwhelming majority of the tested models performed poorly on this task: only DeepSeek V4 achieved a 100% success rate across all four items, while ChatGPT, Gemini, and Claude each failed three of the four. This is a particularly insightful finding given that DeepSeek was the only model to fail the far simpler Axiom D items in Test 3, where the other three models had a perfect record – the model rankings are essentially inverted between the two tasks.

This finding needs one qualification, already flagged in Section **??**: CC-4, though designed as a trivial control, was failed at the same 75% rate as the genuinely hard CC-1 and CC-2, rather than patterning with the easy control CC-3. Because three of the four Task C items happen to have ground truth N and only CC-3 has ground truth Y, a model with a generic acquiescence bias – a default tendency to call a presented statement set consistent – would reproduce the exact accuracy pattern observed without necessarily engaging in the CTD-specific reasoning failure the task was designed

to isolate. The present four-item design cannot distinguish these two accounts, and doing so is listed below as a concrete next step.

Limitations

Several limitations bear directly on how far these results should be read. The item set is small (35 tested items across five batteries, plus 7 further designed-but-untested items in Battery K1/K2), each item was administered once per model rather than resampled, and only four commercial chat products were tested, accessed through their standard consumer interfaces rather than a raw API with fixed decoding parameters – so some of the observed variation may reflect product-level system prompts or safety filtering rather than the base model's reasoning alone. Scoring of the Which? column was performed by hand by a single rater with no independent second scorer, which leaves no measure of inter-rater reliability. The possible acquiescence-bias confound in Task C, discussed above, has not been ruled out. Finally, as noted in the Discussion, LLMs are updated frequently; the specific error rates reported here describe the named model versions as tested in 2026 and are not claimed to generalise to later versions of the same products.

Future Work

The clearest next step is to administer Battery K1 and K2 (Section **??**) to the same four models and fill in their F%/Which? columns, which would close the one research question (RQ2) left unanswered by this report. A second priority is disambiguating the acquiescence-bias account of Task C from the natural-language-coherence account, by adding further balanced controls – for instance, additional short Y-labelled consistent sets that are no less superficially odd-sounding than CC-1/CC-2, so that a pure yes-bias and a genuine CTD-blindness no longer predict the same result. A third direction, already suggested in the Discussion, is repeated sampling per item (multiple runs per model, ideally via the API at a fixed temperature) to determine whether the single-model slips recorded in Tasks A and B (CA-1, CA-2, FB-2, FB-4) are stable failure modes or one-off stochastic noise. Extending the model set beyond four commercial products, and mechanistic follow-up work of the kind called for in the Discussion, would further strengthen any claims about *why* these particular failure patterns occur rather than simply *that* they occur.

Conclusion

The results reported here support a specific picture of deontic reasoning in current LLMs: single-step inferences over paraphrased, negated, or disjunctive obligation and permission statements are handled close to perfectly by three of four tested models, and multi-step contrary-to-duty derivations are handled correctly *when scaffolded*, yet the same models largely fail to recognise that an unscaffolded, self-contained set of everyday obligation statements is internally inconsistent. This is not the pattern that would be predicted by a story in which LLMs have either fully internalised SDL as a formal system or are merely pattern-matching on normative vocabulary. Instead, it suggests that different components of deontic competence (surface-form robustness, guided multi-step derivation, and unprompted consistency-checking) are distinct capacities that do not rise or fall together within or across models.

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